CHANDU PRASAD BHAIRAPU



AI Engineer

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github

ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in machine learning on real-world data, from data pipelines and dataset preparation to model training, fine-tuning and evaluation against ground truth. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by data science work on industrial sensor time-series at Covestro and a strong foundation in Python and PyTorch.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the full data pipeline end-to-end: data ingestion, chunking, prompt engineering and indexing (agentic, deterministic and monolithic strategies), reporting directly to the CTO and working closely with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent extraction algorithms (LangGraph: entity extraction, relationship analysis, consolidation and reflection agents running in parallel) on complex regulatory and technical documents, evaluated against an SAP-system-derived ground truth; precision/recall raised from 33%/45% to 93%/90%.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the shared knowledge store for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a curation web app (React frontend, Python microservices backend) deployed on Azure, where domain experts review, validate and approve knowledge-graph changes before they reach the shared store.
- Used Claude Code, custom skills and agentic coding workflows to ship quickly while keeping to clean code principles.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, Neo4j, MongoDB, Postgres, Vector Search, RAG, MCP, FastAPI, React, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Gathered requirements from business stakeholders on their decision-making workflows; designed and deployed an agentic AI application using LangGraph that routes requests to vector database queries, general inference or summarization.
- Developed a React-based front-end and a Python backend with FastAPI, using Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, following clean code practices; version-controlled in GitLab with CI/CD.
- **Tech stack: Python, LangGraph, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, React, Postman**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated model evaluation of RAG generation and retrieval with RAGAS and PytrecEval, comparing model answers to a chemist-provided ground-truth dataset.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing and fine-tuning models for optimized retrieval.
- Built AWS cloud infrastructure with Lambda, SQS and DynamoDB for PrismaChat RAG evaluation using CDK.

- **Tech stack: Python, PyTorch, Hugging Face Transformers, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, CDK)**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity, reaction progression) from real-world time-series process data, evaluating tree-based vs. deep learning approaches for this tabular/temporal hybrid problem.
- Prepared datasets from raw sensor time-series (temperature, reaction time, process variables): engineered lag features and rolling statistical windows for the tree-based models, while feeding raw sequences directly into the LSTM.
- Benchmarked model performance against traditional empirical curve-fitting methods, improving early identification of batches likely to deviate from target specification.
- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM models for time series analysis of EV load.
- Developed an LSTM that achieved 96% test accuracy with strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Road Damage Dataset & Model Pipeline | PyTorch | DVC | Label Studio | GitHub Actions

- Data pipeline for real-world street images (public RDD2022): shared data store with a metadata catalogue, cleaning, a repeatable labelling workflow in Label Studio and versioned dataset splits with DVC.
- Fine-tunes a pretrained PyTorch computer vision model, evaluates precision/recall per class and recording condition, and reports data gaps that guide the next data collection.

Project: Image Classification | PyTorch: trained and compared CNN, RNN and ResNet models on (permuted) MNIST and Fashion-MNIST, reaching 93% accuracy.

TECHNICAL SKILLSET

- **Machine Learning:** Python, PyTorch, TensorFlow/Keras, Scikit-Learn, XGBoost, Hugging Face Transformers, training and fine-tuning (QLoRA, retrieval fine-tuning), model evaluation, time-series forecasting
- **Data:** data pipelines, data preparation, data quality and feature engineering on sensor data, Pandas, NumPy, SciPy, SQL, Postgres, MongoDB, Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search)
- **Engineering & Cloud:** FastAPI, Docker, Kubernetes, AWS (EC2, Lambda, SQS, CDK), Azure, Git, GitLab CI/CD, Claude Code
- **Languages:** English (Business Fluent), German (B1), Hindi (Fluent), Telugu (Native)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering (Minor in Computer Science & Engineering) — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles